Unit-V

(Solve any two parts)

Q. 5. (a) Define the term shape factor. Derive an expression for the shape factor in case of radiation exchange between two surfaces. **10**

(b) Consider two large parallel plates, one at 1000 K with emissivity 0.8 and other is at 300 K having emissivity 0.6. A radiation shield is placed between them. The shield has emissivity as 0.1 on the side facing hot plate and 0.3 on the side facing cold plate. Calculate percentage reduction in radiation heat transfer as a result of radiation shield. **10**

(c) Write notes on : **10**

(i) Flame radiation

(ii) Solar radiation

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ME/M Tech (1st Semester)
Examination, Nov.-Dec., 2014

Branch : Mechanical Engg.

Specialization : Thermal Engineering

ADVANCED HEAT TRANSFER (OLD)

Time Allowed : Three Hours

Maximum Marks : 100

Minimum Pass Marks : 40

Note : Answer all questions as directed in each unit. Use of HMT data book and steam tables/ psychrometric chart is permitted.

Unit-I

(Solve any two parts)

Q. 1. (a) Starting from first principle derive general heat conduction equation in cylindrical coordinate. **10**

(b) During heat treatment, cylindrical pieces of 25 mm diameter, 30 mm height and at 30°C are placed in a furnace at 750°C with convection coefficient 80 W/m^2-deg. Calculate the time required to heat the pieces to 600°C. What will be the shortfall in temperature if the pieces are taken out from the furnace after 280 seconds. Assume the following property values : **10**

Density = 7850 kg/m^3; Specific heat = 480 J/kg-K, Thermal conductivity = 40 W/m-deg

(c) Describe the procedure for analysis of moving boundaries and moving heat sources problem. **10**

Unit-II

(Solve any two parts)

Q. 2. (a) Carryout the dimensional analysis with the help of Buckingham-π theorem and show that

$$\phi\left(\frac{P_{02}}{P_{01}}, \frac{T_{02}}{T_{01}}, \frac{\dot{m}\sqrt{RT_{01}}}{D^2 P_{01}}, \frac{ND}{\sqrt{RT_{01}}}\right) = 0$$

where P_{02}, P_{01} represent the pressure, T_{02}, T_{01} is temperature, $\dot{m}$ is mass flow rate, N is r.p.m. and D is diameter and R is characteristic gas constant. **10**

(b) Derive the Von Karman Momentum equation in the integral form for hydrodynamic boundary layer **10**

$$\tau_w = \rho U^2 \frac{\partial}{\partial x}\left[\int_0^{\delta}\left(1-\frac{u}{U_\infty}\right)\frac{u}{U_\infty}\,dy\right]$$

(c) Air at 20°C enters a circular pipe 16 cm in diameter and 160 cm long whose wall is maintained at 160°C. How much heat is transfered ? Assume the flow is turbulent and same heat transfer coefficient throughout. **10**

Average velocity of air = 15 m/s

Use Reynold analogy.

Take the following properties at average temperature of air at 90°C.

$v = 22 \times 10^{-6}$ m²/sec, K = 0.027 W/m-K

Unit-III

(Solve any two parts)

Q. 3. (a) A rectangular duct 30 cm × 20 cm in cross-section carries cold air. The temperature of the outer surface of the duct is 5°C and

surrounding air temperature is 25°C. Find the heat gain by the duct assuming one metre duct is exposed to the air in vertical position. The properties of air at $\left(\frac{25+5}{2}\right)$ = 15°C are as : **10**

$\rho = 1.2$ kg/m², $\mu = 18.17 \times 10^{-6}$ kg/m-s

$v = 14.61 \times 10^{-6}$ m²/sec, $C_p = 1006$ J/kg-K

K = 0.0256 W/m-K

(b) Define and explain the following with example : **10**

(i) Combined free and forced convection

(ii) Combined convection and radiation

(c) 50 kg of water per minute is heated from 30°C to 50°C by passing through a pipe of 2 cm diameter. The pipe is heated by

condensing the steam on its surface at 100°C. Find the length of the pipe required. Take the following properties of water at mean temperature of 90°C. **10**

ρ = 965 kg/m^3, K = 0.585 W/m-K

C_p = 4200 J/kg-K, v = 0.33 × 10^{-6} m^2/sec.

Unit-IV

(Solve any two parts)

Q. 4. (a) With the help of diagram explain the construction and working of heat pipe. List some applications and limitations of the heat pipe. **10**

(b) Explain the following : **10**

(i) Nucleate boiling

(ii) Pool boiling

(iii) Film boiling

(iv) Forced convection boiling

(v) Burnout point

(c) A vertical tube of 60 mm outside diameter and 1.2 m long is exposed to steam at atmospheric pressure. The outer surface of the tube is maintained at a temperature of 50°C by circulating cold water through the tube. Calculate the following : **10**

(i) The rate of heat transfer to the coolant, and

(ii) The rate of condensation of steam.